Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	BLUFFING
Project / Solution Name	Domain 1
Name (Leader)	Ivannov Tatang Kennedy
Email & Phone (Leader)	ivannovkennedy@gmail.com (+60177414536)
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://youtu.be/ajJs44XIqno?si=Xx8YH0mSdkSs6P7E
Code Repository URL (public / access granted to judges)	https://github.com/Ivannov-arch/telos-sdc

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Telos uses a Next.js frontend hosted on Replit, connected to a FastAPI backend written in Python through REST APIs. The backend processes user profiles, matches skills with suitable occupations, and retrieves relevant career information from Supabase PostgreSQL with pgvector.

ESCO provides standardised skills and occupation data, while O*NET supplies occupational skills and tasks. OpenDOSM Job Market Insights provides Malaysian job-market signals, and the UN SDG API supplies official SDG information for career mapping.

For AI processing, RunPod Serverless GPU hosts BGE-M3 for semantic skill matching and a language model for structured AI outputs. Supabase stores user profiles, occupation data, embeddings and curated SDG mappings. The results are returned to the frontend and displayed through the career dashboard.

This modular architecture allows the team to develop and test core features first, then expand the system with updated job-market analysis, personalised reskilling pathways and career-disruption scenarios.

User Interface (Next.js/Replit) → Backend (FastAPI/Python) → AI Processing (RunPod: BGE-M3 + LLM) → Database (Supabase PostgreSQL + pgvector) → External Data Sources (ESCO, O*NET, OpenDOSM and UN SDG API) → Career Results Dashboard.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	**Purpose in Project**	**License**
Next.js 15.5	Builds the frontend and career dashboard.	MIT License
Python 3.12	Runs the backend logic.	PSF License
FastAPI 0.115	Connects the frontend to backend APIs.	MIT License
PostgreSQL 17	Stores user profiles, skills and occupation data.	PostgreSQL License
pgvector 0.8.0	Enables vector storage and similarity earches for career matching.	PostgreSQL License

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
ESCO API	Retrieves skills and occupation classifications.	free
UN SDG API	Retrieves official SDG goals and targets.	free
OpenDOSM Job Market Insights	Provides Malaysian job-market data.	free
RunPod Serverless Endpoint	Runs BGE-M3 embeddings and LLM inference.	sponsor-provided
Supabase API / SDK	Connects the backend to stored career and skills data.	free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Hosts the Next.js frontend and FastAPI backend.	Replit deployment plan used
RunPod Serverless GPU	Runs AI embeddings and language-model inference.	Serverless; credits / paid
Supabase	Hosts PostgreSQL database and pgvector for career matching.	Free tier

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
BGE-M3	BAAI	Pre-trained; generates embeddings for semantic career and skills matching.	MIT License
Llama 3.2B	Meta	Pre-trained; generates structured career recommendations and explanations	Llama 3.2 Community License

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
ESCO Skills & Occupations	https://esco.ec.europa.eu/en/use-esco/download	Public	CC BY 4.0	No
O*NET Database	https://www.onetcenter.org/database.html	Public	Public Domain (US government)	No

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
OpenDOSM Job Market Insights	https://open.dosm.gov.my/data-catalogue	Public	Malaysian Government Open Data	No
MASCO 2020	https://emasco.mohr.gov.my/intro	Public	Malaysian Government Publication	No
UN SDG Data	https://unstats.un.org/sdgs/UNSDGAPIV5/swagger/index.html	Public	UN Terms of Use	No

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Gemini 3.8-flash	Research and coding assistance	moderate
Gemini Pro	Research and information analysis	minor
Gemini Deep Research	In-depth research and source gathering	minor
Perplexity	Finding information and research sources	minor
Claude Sonnet 4.6	Coding assistance, debugging and code generation	moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team has tested AI-generated career recommendations using synthetic user profiles and compare the suggested roles against ESCO and O*NET skill requirements. We checked whether match scores reflect users' existing skills, whether identified skill gaps are accurate, and whether SDG mappings align with official UN targets. AI-generated outputs will be validated against stored data to reduce fabricated careers, skills or SDG links. We also tested structured JSON outputs to ensure they follow the required format. We will vigorously review the results, correct errors and remain responsible for the final recommendations presented by ***Telos***.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
RunPod	AI model hosting platform / GPU credits	Used to host BGE-M3 embeddings and the selected language model for career matching and structured AI outputs.
Replit	App development and deployment platform	Used to build and deploy the Telos web application, including the Next.js frontend and FastAPI backend.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

RunPod supports AI inference and embedding generation, while Replit supports application development and deployment. Both are useful to the current implementation, but they could be replaced by alternative model-hosting and deployment platforms with appropriate modifications. The core functionality of Telos depends on career matching, skills-gap analysis and SDG mapping rather than on any single provider.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Telos collects only non-sensitive professional attributes: extracted skill keywords, years of experience, and target roles submitted via structured forms or uploaded resume PDFs. We do not harvest, store, or process government IDs, contact info, home addresses, or financial data.

Uploaded CVs are processed in-memory via pdfplumber to extract skill entities, and raw text is immediately discarded or stored ephemerally. To generate embeddings, anonymized skill tokens are transmitted over TLS 1.3 encrypted HTTPS exclusively to our private, self-hosted RunPod GPU container (/embed endpoint); no user data is ever sent to public commercial LLM APIs (e.g., OpenAI or Anthropic).

Data is stored in Supabase PostgreSQL under strict Row-Level Security (RLS) policies with service-role authorization. Demonstration session data is purged within 30 days post-hackathon evaluation, and users can request immediate profile deletion.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

1. **AI Exposure Bias**
AIOE and GPTs-are-GPTs may overestimate job replaceability due to cognitive-task bias (Rai, 2026). Mitigation: Weight Webb scores at 50% and interpret results as task-compression signals, not job-loss probabilities.
2. **AI Hallucinations**
LLMs may generate inaccurate skills or SDG links. Mitigation: Ground recommendations in ESCO, O*NET and official UN data, using deterministic skill-gap calculations and curated SDG mappings.
3. **RunPod Dependency**
GPU cold starts or service outages may delay AI processing.

Mitigation that we have implemented is using a fallback skill-overlap scoring method when embedding services are unavailable.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



24 October 2026